

VOLUME IV - REFERENCES / STANDARDS

# Standards Index

Technical Standards Applicable to Equipment, Communications, Safety and Test

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|---------------------|----------------------------------------------------------------------------------|
| Document ID         | NMDC-LVH-RF-03                                                                   |
| Volume              | IV - References                                                                  |
| Programme           | Safe & Continuous Haulage under Low-Visibility Monsoon Conditions                |
| Site of Application | Bailadila Iron Ore Mines - Kirandul & Bacheli Complexes, Dantewada, Chhattisgarh |
| Classification      | Internal Research / Solution Design Input                                        |
| Revision            | v1.0                                                                             |
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ABSTRACT

An index of the technical standards families relevant to specifying, qualifying, testing and assuring the proposed system, covering environmental protection and qualification, earth-moving machinery safety, functional safety, communications and V2X, positioning, visibility measurement, human-machine interface, cybersecurity and quality. Each family entry states what it governs, how it applies here, and what the programme must do with it. The document closes with a procurement specification checklist and guidance on the difference between citing a standard and complying with one.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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# 1. How to Use This Index

This index identifies **standards families** rather than pinned clause numbers. Standards are revised, withdrawn and superseded; a specification that cites a stale edition creates a compliance argument rather than settling one.

**The rule the programme should adopt:** a procurement specification cites a standard **by number, by current edition and by the specific clause or severity level required**, and requires the supplier to furnish a test report or certificate from a competent body. A specification that says "shall comply with applicable standards" has specified nothing. Every entry in this index should become a line in the compliance register with a named edition, a named severity and a named evidence artefact.

Entries carry a code ST-nn and record: **Family, Governs, Application here, and Programme action.**

## 2. Ingress, Environmental and Vehicular Qualification

**ST-01. IEC 60529 - Degrees of protection provided by enclosures (IP code).** *Governs:* the IP rating system for solid and liquid ingress. *Application:* requirement ER-01 sets IP67 as the minimum for externally mounted equipment and IP69K where pressure washing applies. Note that IP69K - high-pressure, high-temperature water jets - is defined in the road-vehicle standards lineage rather than in IEC 60529 itself, and must be cited accordingly. *Action:* specify IP67 or IP69K explicitly per mounting location; require test reports, not declarations.

**ST-02. ISO 20653 - Road vehicles, degrees of protection (IP code), protection against foreign objects, water and access.** *Governs:* the vehicular IP framework including IP6K9K. *Application:* the correct citation basis for pressure-wash protection on HEMM-mounted equipment. *Action:* cite this rather than IEC 60529 alone where washing is a factor.

**ST-03. IEC 60068 series - Environmental testing.** *Governs:* standard test methods for cold, dry heat, damp heat (steady state and cyclic), vibration, shock, salt mist and combined stresses. *Application:* the evidence basis for requirements ER-02 (-5 to +65 deg C ambient, higher at soak positions), ER-03 (95 per cent RH with condensation cycling) and ER-04 (HEMM shock and vibration). Damp heat cyclic testing is the one that matters most for monsoon conditions, because it is condensation cycling rather than steady humidity that kills electronics. *Action:* specify the exact test, duration and severity for each; require reports.

**ST-04. ISO 16750 series - Road vehicles, environmental conditions and testing for electrical and electronic equipment.** *Governs:* electrical loads (including load dump, reverse polarity, overvoltage, transients), mechanical loads, climatic loads and chemical loads for vehicle-mounted electronics. *Application:* the natural basis for requirement ER-05 on wide-range 24 V DC input with load-dump and transient protection. *Action:* specify the applicable parts and the code letters for the mounting location.

**ST-05. ISO 7637 series - Road vehicles, electrical disturbances from conduction and coupling.** *Governs:* conducted transient immunity and emission on vehicle supply lines. *Application:* ER-05. *Action:* specify pulse types and severity levels.

**ST-06. CISPR 25 / IEC 61000 series - Electromagnetic compatibility.** *Governs:* radiated and conducted emissions and immunity for vehicle components (CISPR 25) and general EMC immunity and emission (IEC 61000). *Application:* requirement ER-09, demonstrated compatibility with existing site systems, **including explicit verification against blast initiation systems.** *Action:* specify EMC compliance and, separately, require a site-specific coexistence test with the blasting authority - a generic EMC certificate does not discharge this obligation.

**ST-07. IEC 62305 / IEC 61643 series - Lightning protection and surge protective devices.** *Governs:* lightning protection system design and surge protection on power and signal lines. *Application:* requirement ER-06 for exposed ridge infrastructure - RTK bases, repeaters, visibility nodes. *Action:* specify surge protection class for all outdoor cabling and antenna feeds.

**ST-08. Relevant BIS standards for enclosures, electrical safety and photovoltaic systems.** *Application:* Indian conformity for the solar-powered roadside nodes of requirement ER-08. *Action:* confirm the current BIS requirements for the PV and battery elements.

### 3. Earth-Moving Machinery Safety

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**ST-09. ISO 3450 - Earth-moving machinery, wheeled or high-speed rubber-tracked machines, braking system performance requirements and test procedures.** *Governs:* service, secondary and parking brake performance for earth-moving machinery. *Application:* **directly relevant to the stopping-distance derivation of Volume I Document 3, Section 3.1.** That derivation adopted an indicative effective deceleration coefficient of 0.25 on a wet ore-fines surface with a 1.5 s perception-reaction time. Those are conservative placeholders. *Action:* **replace the indicative parameters with measured braking performance for the actual Bailadila fleet, referenced to this standard's test basis.** The entire detection-range requirement PR-01 depends on them, so this is not an academic refinement - if real deceleration is materially lower than 0.25 on wet ore fines, the required detection range rises.

**ST-10. ISO 5006 - Earth-moving machinery, operator's field of view, test method and performance criteria.** *Governs:* the assessment of operator visibility and blind zones. *Application:* defines the blind-zone geometry that requirement FR-13 and PR-03 must cover, and provides the objective basis for specifying short-range sensor coverage per machine class. *Action:* obtain or conduct ISO 5006 visibility assessments for each Bailadila machine class; use them to derive sensor placement rather than placing sensors by inspection.

**ST-11. ISO 16001 - Earth-moving machinery, object detection systems and visibility aids, performance requirements and tests.** *Governs:* **the directly applicable standard for object detection systems on earth-moving machinery**, including performance requirements and test methods for detection zones and visibility aids. *Application:* the natural specification and test basis for the short-range and blind-zone layers of the proposed system, and a credible reference point in any DGMS discussion. *Action:* **obtain the current edition and adopt it as the baseline test basis for the object detection layer**, supplementing it with the MOR-referenced testing of requirement AR-02, which it does not cover.

**ST-12. ISO 6165 - Earth-moving machinery, basic types, identification, terms and definitions.** *Application:* consistent machine classification in the V2V message set of Volume III Document 4, Section 2.1.

**ST-13. ISO 3471 (ROPS) and ISO 3449 (FOPS).** *Governs:* roll-over and falling-object protective structures. *Application:* no mounting, harness routing or modification for the new system may compromise a certified protective structure. *Action:* **an explicit constraint on installation design.** Drilling or welding to a ROPS structure invalidates its certification. Installation procedures must state this.

**ST-14. ISO 6405 - Earth-moving machinery, symbols for operator controls and other displays.** *Application:* requirement HR-02's preference for non-textual interfaces is better served by adopting established machinery symbology than by inventing new icons. *Action:* align interface symbology with the standard where applicable symbols exist.

### 4. Functional Safety

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**ST-15. IEC 61508 - Functional safety of electrical/electronic/programmable electronic safety-related systems.** *Governs:* the generic functional safety lifecycle, Safety Integrity Levels, and the requirements for hardware and systematic capability. *Application:* the appropriate framework for the safety argument of an industrial system of this kind. *Action:* determine, early, whether NMDC and DGMS expect a formal functional safety case. This materially affects cost, schedule and supplier selection and must not be discovered late.

**ST-16. ISO 26262 - Road vehicles, functional safety.** *Governs:* the automotive functional safety lifecycle and ASIL classification. *Application:* many of the components selected in this package - automotive radar, automotive-grade SoCs, GNSS/INS units - are developed to ISO 26262. *Action:* require suppliers to state the ASIL capability of components supplied. **Note carefully:** component ASIL capability does not confer safety integrity on the integrated system. An assembly of ASIL-B components integrated without a safety case is not a safety-rated system. This distinction should be stated in any tender to prevent suppliers claiming otherwise.

**ST-17. ISO 13849 / IEC 62061 - Safety of machinery, safety-related control systems.** *Governs:* performance levels and safety integrity for machinery control systems. *Application:* **becomes directly relevant if and when intervention functions (requirement FR-26, Phase 4) are pursued**, since those take control authority over the machine. *Action:* not required for the advisory system; scope it into the Phase 4 approval planning.

**ST-18. ISO 17757 - Earth-moving machinery and mining, autonomous and semi-autonomous machine system safety.** *Governs:* **the directly applicable standard for autonomous and semi-autonomous machine systems in**

**earth-moving and mining**, covering safety requirements, operating zones, and the interaction of autonomous machines with people and other machines. *Application*: highly relevant if autonomy or intervention is pursued in Phase 4, and a useful reference framework even for the advisory system because it addresses mixed autonomous-and-manned operation - architecturally the same problem as mixed equipped-and-unequipped fleet operation (failure mode F4). *Action*: obtain and review; it is the most directly applicable international standard to the programme's long-term direction.

## 5. Radar and Radio Equipment

- ST-19. ETSI EN 301 091 and related short-range device standards for 76-81 GHz radar equipment.** *Governs*: harmonised requirements for radar equipment in the automotive millimetre-wave bands. *Application*: the technical basis on which candidate radar equipment is manufactured, even where Indian authorisation is the governing legal requirement. *Action*: require suppliers to evidence compliance, and separately confirm the Indian spectrum position per Volume IV, NMDC-LVH-RF-02, entry GR-E1.
- ST-20. ETSI EN 302 264 and related standards for 77-81 GHz short-range radar.** *Application*: as above for the mid- and short-range sensors.
- ST-21. Indian type approval and equipment authorisation requirements for radio equipment.** *Application*: requirement GR-04. *Action*: verify the supplier **holds** current approval; do not accept a statement that approval is obtainable.
- ST-22. IEC 62368-1 - Audio/video, information and communication technology equipment, safety requirements.** *Application*: general electrical safety for in-cab display and compute equipment.

## 6. V2X Communications

- ST-23. IEEE 802.11p and the ETSI ITS-G5 standards family (EN 302 663 and associated).** *Governs*: the DSRC/ITS-G5 access layer and European ITS station architecture. *Application*: one of the two candidate V2V bearers evaluated in Volume III Document 4, Section 3.3.
- ST-24. 3GPP specifications for C-V2X - LTE-V2X (Release 14/15) and NR-V2X (Release 16 onward), PC5 sidelink Modes 4 and 2.** *Governs*: the direct device-to-device sidelink that is the package's preferred bearer, subject to the RF survey. *Application*: Volume III Document 4, Sections 3.2 and 3.5. *Action*: specify the release and mode required; note that NR-V2X unicast and groupcast with feedback enables acknowledged delivery for critical messages such as movement authority.
- ST-25. SAE J2735 - V2X communications message set dictionary; and SAE J2945 series - performance requirements for V2X safety communications.** *Governs*: standardised message definitions including the Basic Safety Message, and minimum performance requirements for broadcast rate, accuracy and latency. *Application*: the proposed message set of Volume III Document 4, Section 2 should **extend** these standard definitions rather than replace them, preserving interoperability and reusing a well-considered baseline. The package's additions - position confidence, system health and effective sensing range, road segment and lane, operating state - are extensions, and should be documented as such. *Action*: adopt the standard message set as the foundation; specify extensions explicitly.
- ST-26. ETSI TS 102 894 and the Cooperative Awareness Message / Decentralized Environmental Notification Message family.** *Application*: the European analogue to SAE J2735; the DENM concept is the direct precedent for the Hazard Marker message that this package identifies as the **only** mechanism addressing the washout hazard.
- ST-27. ETSI TS 102 687 - Decentralized congestion control mechanisms for intelligent transport systems.** *Application*: directly relevant to the scalability analysis of Volume III Document 4, Section 5, at the 60-vehicle design density of requirement CR-03. *Action*: specify congestion control conformance; require loaded-channel test evidence rather than analytical estimates.
- ST-28. IEEE 1609 series - Wireless Access in Vehicular Environments (WAVE), including 1609.2 security services.** *Governs*: the vehicular networking stack and its security services, including message signing and certificate management. *Application*: the mature security framework referenced in Volume III Document 4, Section 8 - adopt rather than invent. *Action*: base the site PKI design on this framework.

## 7. Positioning and Geodesy

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**ST-29. RTCM SC-104 - Differential GNSS correction message standards.** *Governs:* the standard format for RTK and DGNSS corrections. *Application:* interoperability between the RTK base network and vehicle receivers; avoids vendor lock-in at the correction interface, supporting requirement IR-01. *Action:* specify RTCM correction output and input explicitly.

**ST-30. NMEA 0183 and related positioning output formats.** *Application:* standard position output interfaces.

**ST-31. Indian geodetic datum and survey practice; and the mine survey datum in use at Bailadila.** *Application:* requirement FR-05 references position to the mine survey datum. The digital road model, RTK base coordinates and all vehicle positions must share one datum and one transformation. *Action:* **settle the datum and transformation question explicitly and early.** Datum inconsistency between the survey department, the road model and the positioning system produces a systematic offset that is invisible in testing and dangerous in operation - the model would draw the road consistently displaced from where it is.

**ST-32. ICAO and RTCA standards practice for GNSS integrity and protection levels.** *Application:* the source of the protection-level concept adopted in Volume III Document 2, Section 1.8. Not directly applicable as a compliance standard to a mining system, but the correct methodological reference for the integrity argument.

## 8. Visibility and Meteorological Measurement

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**ST-33. WMO Guide to Instruments and Methods of Observation (WMO-No. 8), and ICAO Annex 3 provisions on Runway Visual Range.** *Governs:* the definition of meteorological optical range, instrument siting and exposure, measurement uncertainty, and the reference practice for visibility measurement. *Application:* the methodological basis for the visibility network of requirements FR-01, FR-02 and, critically, for the calibrated independent measurement mandated by requirement AR-02. *Action:* **specify the visibility network against WMO practice, including siting, exposure and calibration traceability.** The credibility of the entire acceptance regime rests on the measurement being defensible.

**ST-34. ISO 28902 series - Air quality, environmental meteorology, and optical remote sensing methods including visibility measurement.** *Application:* supplementary methodological reference for forward-scatter and transmissometer instruments.

**ST-35. Instrument-level standards and calibration traceability for transmissometers and forward-scatter visibility meters.** *Action:* require calibration traceable to a recognised standard, with a stated calibration interval, per the maintenance schedule in Volume III Document 2, Section 9.

## 9. Human-Machine Interface

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**ST-36. ISO 15008 - Road vehicles, ergonomic aspects of transport information and control systems, specifications and test procedures for in-vehicle visual presentation.** *Governs:* legibility, character size, contrast and luminance for in-vehicle displays. *Application:* requirements HR-01, HR-07 and HR-08 on glanceable, luminance-adaptive display. *Action:* specify conformance; it provides objective acceptance criteria for what would otherwise be a subjective assessment.

**ST-37. ISO 15005 and ISO 17287 - Dialogue management principles and suitability assessment for transport information and control systems while driving.** *Application:* requirement HR-09 that the interface require no interaction during normal driving, and the general principle that the system must not add task load.

**ST-38. ISO 15006 - Specifications and test procedures for auditory presentation in road vehicles.** *Application:* requirement HR-04 on audible alerts distinguishable in an 85 dB(A) cab ambient with hearing protection in use.

**ST-39. ISO 9241 series - Ergonomics of human-system interaction.** *Application:* general usability principles for the control room interface of requirements OR-01 to OR-09, which is a supervisory rather than an in-vehicle interface and is therefore governed by different ergonomics.

**ST-40. Accessibility and colour-vision guidance.** *Application:* requirement HR-03 that colour not be the sole carrier of safety-critical information.

## 10. Cybersecurity

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- ST-41. IEC 62443 series - Industrial communication networks, security for industrial automation and control systems.** *Governs:* security levels, zones and conduits, and lifecycle requirements for industrial control system security. *Application:* the appropriate framework for the site-wide system, covering the control room, backhaul and roadside infrastructure. *Action:* determine the target security level in Phase 0; it affects architecture, not just configuration.
- ST-42. ISO/SAE 21434 - Road vehicles, cybersecurity engineering.** *Governs:* automotive cybersecurity lifecycle and risk assessment. *Application:* the vehicle-side equipment, and the framework most suppliers of automotive-derived components will already work within. *Action:* require suppliers to evidence a cybersecurity engineering process.
- ST-43. IEEE 1609.2 (see ST-28) - V2X security services.** *Application:* message authentication per requirement CR-04.
- ST-44. ISO/IEC 27001 - Information security management systems.** *Application:* the organisational frame for the data governance requirement IR-04, particularly the handling of operator-attributable data. *Action:* the data governance policy required before deployment should be written within an established ISMS frame rather than as a standalone document.

## 11. Data, Interfaces and Integration

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- ST-45. ISO 11898 (CAN) and CAN FD; SAE J1939 - vehicle network protocols.** *Application:* the machine CAN interface of Volume III Document 2, Section 7. *Action:* specify **read-only** access and galvanic isolation per requirement ER-05. Write access to a machine network is a different regulatory category and must not be introduced incidentally.
- ST-46. IEEE 802.3 automotive Ethernet variants.** *Application:* the sensor-to-compute interface for imaging radar, whose data rate exceeds CAN capability.
- ST-47. Open geospatial and mine-modelling data format practice.** *Application:* the digital road model of requirement FR-06 should be held in an open, documented format to satisfy IR-01 and to allow the mine survey department to maintain it with its own tools. *Action:* **do not accept a proprietary road model format.** The model must be maintainable by NMDC independently of the system supplier, or requirement FR-07 (field update within a shift without vendor involvement) cannot be met in practice.
- ST-48. IEEE 1588 Precision Time Protocol; GNSS time as a synchronisation source.** *Application:* requirement IR-03, with the sub-millisecond target derived in Volume III Document 4, Section 9.

## 12. Quality and Assurance

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- ST-49. ISO 9001 - Quality management systems.** *Application:* supplier qualification baseline.
- ST-50. ISO 45001 - Occupational health and safety management systems.** *Application:* the management frame into which the new controls, revised procedures and training are integrated.
- ST-51. ISO/IEC 17025 - General requirements for the competence of testing and calibration laboratories.** *Application:* **requirement AR-07 and AR-02.** Test reports and calibration certificates should come from competent bodies. This matters most for the visibility instrumentation, because the entire MOR-referenced acceptance argument rests on the visibility measurement being defensible.
- ST-52. ISO 31000 - Risk management.** *Application:* the frame for the risk register of Volume II Document 3, Section 8 and for the risk assessments required under requirement GR-05 where operating procedures change.

## 13. Procurement Specification Checklist

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A tender for this system should require, at minimum, the following evidence. Each line maps to an index entry above.

| #  | Evidence required                                                                                                            | Reference                                                                   |
|----|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1  | IP rating test report per mounting location, IP67 or IP6K9K                                                                  | ST-01, ST-02                                                                |
| 2  | IEC 60068 environmental test reports: cold, dry heat, damp heat cyclic, vibration, shock - with stated severities            | ST-03                                                                       |
| 3  | ISO 16750 and ISO 7637 electrical qualification reports for the stated mounting code                                         | ST-04, ST-05                                                                |
| 4  | CISPR 25 / IEC 61000 EMC reports <b>plus</b> a site coexistence test with blast initiation systems                           | ST-06                                                                       |
| 5  | Surge and lightning protection design for all outdoor assets                                                                 | ST-07                                                                       |
| 6  | ISO 16001 object detection system performance test report                                                                    | ST-11                                                                       |
| 7  | ISO 5006 field-of-view assessment per machine class, with derived sensor coverage                                            | ST-10                                                                       |
| 8  | Confirmation that installation does not modify ROPS/FOPS structures                                                          | ST-13                                                                       |
| 9  | Component ASIL capability statement, with an explicit statement of what it does and does not imply for the integrated system | ST-16                                                                       |
| 10 | Radar equipment compliance evidence and <b>current Indian type approval held by the supplier</b>                             | ST-19, ST-20, ST-21                                                         |
| 11 | V2X standard conformance, release and mode stated; message set documented as extensions to the standard dictionary           | ST-24, ST-25                                                                |
| 12 | Congestion control conformance with <b>loaded-channel test evidence at 60-vehicle density</b>                                | ST-27                                                                       |
| 13 | V2X security architecture based on an established framework; PKI design                                                      | ST-28, ST-43                                                                |
| 14 | RTCM-standard correction interfaces, input and output                                                                        | ST-29                                                                       |
| 15 | Datum and transformation definition agreed with NMDC survey                                                                  | ST-31                                                                       |
| 16 | Visibility instrumentation specified to WMO practice with traceable calibration                                              | ST-33, ST-35, ST-51                                                         |
| 17 | ISO 15008 / 15006 display and auditory conformance                                                                           | ST-36, ST-38                                                                |
| 18 | Cybersecurity engineering evidence                                                                                           | ST-41, ST-42                                                                |
| 19 | <b>Open, documented road model format, maintainable by NMDC without the supplier</b>                                         | ST-47, FR-07                                                                |
| 20 | Read-only, galvanically isolated machine network interface                                                                   | ST-45                                                                       |
| 21 | <b>Detection performance stated and demonstrated against calibrated MOR down to 5 m</b>                                      | AR-02, AR-03 - <i>no standard covers this; it must be specified bespoke</i> |
| 22 | <b>Measured false alert rate over not less than 200 vehicle-operating-hours</b>                                              | HR-06, AR-04 - <i>no standard covers this either</i>                        |

Items 21 and 22 are the two most important lines in the checklist and the two for which no standard exists. Every other line can be discharged by a certificate. These two require the supplier to do something the market does not currently do, and they are precisely the requirements that distinguish a system that will work at Bailadila from one that will demonstrate well on a clear day.

## 14. Citing Versus Complying

Three cautions, drawn from the general experience of technology procurement in heavy industry.

**A standard cited without an edition, clause and severity has not been specified.** "Complies with IEC 60068" is meaningless - the family contains dozens of tests at many severities. Specify the test, the severity and the duration.

**Component compliance is not system compliance.** A system assembled from ISO 26262 ASIL-B components is not an ASIL-B system. A radar with an ETSI compliance certificate does not make the installed multi-sensor system compliant with anything. Suppliers will, in good faith, present component certificates as system evidence. The specification should pre-empt this by requiring an explicit statement of what each certificate covers.

**No standard covers the thing that matters most here.** There is no standard for detection performance at a defined meteorological optical range on a mining haul road. ISO 16001 covers object detection zones but not visibility conditions. The WMO guidance covers measuring visibility but not sensing through it. **The programme must therefore write its own test specification for items 21 and 22 of the checklist**, using ISO 16001 for the detection methodology, WMO practice for the visibility measurement, and ISO/IEC 17025 for the competence of whoever performs it. That bespoke test specification is, in the assessment of this package, one of the more valuable documents the programme will produce - and it is likely to be of interest to DGMS and to the wider Indian mining industry, because the gap it fills is market-wide.